

Table 15: Average character index of where the answer span begins in a generated response for each dataset and model pair for the few-shot direct answer setting. We use these numbers as a proxy for the model following instructions (i.e. generating reasoning before an answer). Prompt modifications were made to ensure CoT prompts resulted in longer generations and direct answer prompts led to short generations.

Few-shot Direct Answer Span Location By Character Index										
dataset	Meta-Llama 2.7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2.9b	Phi-3 Small 8k	Qwen 2.7b	Qwen 2.72b	GPT-4o Mini	Gemini 1.5 Flash
CommonsenseQA	87	8	27	8	8	8	10	8	8	8
AGIEval LSAT LR	25	24	24	24	24	24	24	24	31	24
AGIEval LSAT AR	25	24	24	24	24	24	24	24	27	24
AGIEval LSAT RC	25	24	24	24	24	24	24	24	25	24
ContextHub Deductive L1	19	19	19	19	19	19	19	19	19	19
ContextHub Deductive L2	19	19	19	19	19	19	19	19	19	19
ContextHub Abductive L1	19	19	19	19	19	19	19	19	19	19
ContextHub Abductive L2	19	19	19	19	19	19	19	19	19	19
MuSR Murder Mysteries	-1	8	27	8	8	8	8	8	8	8
MuSR Team Allocations	-1	21	19	19	27	21	27	23	8	8
MuSR Object Placements	-1	8	27	8	8	8	8	8	8	8
MMLU	19	18	19	19	19	18	18	18	19	19
MMLU Pro	19	19	38	19	20	20	19	19	19	19
GPQA	-1	19	19	19	19	19	19	19	19	19
MATH	29	36	29	29	28	30	30	41	28	28
GSM8k	22	23	23	22	22	23	22	24	27	28

Table 16: Average character index of where the answer span begins in a generated response for each dataset and model pair for the few-shot CoT setting. We use these numbers as a proxy for the model following instructions (i.e. generating reasoning before an answer). Prompt modifications were made to ensure CoT prompts resulted in longer generations and direct answer prompts led to short generations.

Few-shot CoT Answer Span Location By Character Index										
dataset	Meta-Llama 2.7b	Mistral 7b	Meta-Llama 3.1 8b	Meta-Llama 3.1 70b	Gemma 2.9b	Phi-3 Small 8k	Qwen 2.7b	Qwen 2.72b	GPT-4o Mini	Gemini 1.5 Flash
CommonsenseQA	301	195	470	921	145	192	280	174	219	158
AGIEval LSAT LR	1037	510	464	539	437	359	530	599	894	523
AGIEval LSAT AR	1024	1247	886	768	573	1025	750	835	1033	670
AGIEval LSAT RC	799	378	131	206	164	111	241	205	1086	266
ContextHub Deductive L1	383	386	406	376	359	376	388	364	416	366
ContextHub Deductive L2	736	767	829	822	823	855	612	807	884	809
ContextHub Abductive L1	301	386	428	450	431	413	541	447	575	379
ContextHub Abductive L2	709	586	967	754	804	784	829	821	905	815
MuSR Murder Mysteries	-1	1280	1693	1702	1225	1338	1246	1719	1974	1419
MuSR Team Allocations	-1	2195	2087	2160	1628	1755	2181	2156	2632	1841
MuSR Object Placements	-1	907	1104	1213	706	919	676	963	1351	853
MMLU	282	266	333	245	265	260	267	243	392	218
MMLU Pro	429	397	424	411	516	425	541	325	681	396
GPQA	-1	848	782	774	615	711	662	703	670	594
MATH	630	705	584	640	747	529	1074	848	1261	553
GSM8k	374	332	352	352	398	372	415	341	651	314

Table 17: The top 3 slices benefiting the most from CoT across MMLU and MMLU Pro for Llama 3.1 8b and 70b. 6 out of 12 of these top slices directly contain “math” or “mathematics.” We dive deeper into each category subsequently and observe that the questions leading to improvements in the other categories are mathematical in nature as well.

Model	Subject	MMLU				Subject	MMLU Pro			
		Direct (%)	CoT (%)	Err. Red. (%)	N		Direct (%)	CoT (%)	Err. Red. (%)	N
Llama 3.1 8b	elementary_mathematics	46.8	88.4	78.1	378	math	23.6	44.8	27.8	1350
Llama 3.1 8b	high_school_mathematics	39.6	71.5	52.8	270	business	29.4	45.6	23.0	789
Llama 3.1 8b	miscellaneous	83.9	89.9	37.3	783	physics	27.9	41.4	18.8	1299
Llama 3.1 70b	elementary_mathematics	82.3	94.7	70.1	378	math	44.5	68.3	42.9	1351
Llama 3.1 70b	medical_genetics	93.0	97.0	57.1	100	business	44.0	67.8	42.5	789
Llama 3.1 70b	high_school_mathematics	61.5	82.2	53.8	270	chemistry	40.5	64.0	39.6	1132

level. In doing so, we can test our hypotheses with much finer granularity than possible by relying on subjective groupings into tasks and categories.

Breakdown by the presence of equations We aim to design an instance-level classifier to determine if CoT is expected to help on a question or not. That is, we want a function $g : \mathbf{q} \rightarrow \{0, 1\}$ where $g(\mathbf{q})$ returns 1 if $\text{extract}(\tilde{\mathbf{y}}_{cot}) = \mathbf{y}^*$ and $\text{extract}(\tilde{\mathbf{y}}_{da}) \neq \mathbf{y}^*$ where $\mathbf{y}^*$ is the gold answer to $\mathbf{q}$. We explored different forms of g ; however, we ultimately found it most effective to use a classifier $g : (\mathbf{q}, \tilde{\mathbf{y}}_{cot}) \rightarrow \{0, 1\}$ which also consults the chain-of-thought produced by the model. This allows us to featurize how the LM solves the problem, particularly whether it uses symbolic reasoning or not.

We find that g can be implemented with a **single feature**: does $\mathbf{q}$ or $\tilde{\mathbf{y}}_{cot}$ contain a “=”? The “=” token very strongly indicates the presence of equations in the problem or its solution, which turn out to be a strong hallmark of symbolic reasoning.⁶

We plot the overall CoT delta (performance of CoT minus the performance of direct answer) for both MMLU and MMLU Pro across multiple models between two bins according to this classifier g , labeled as “With =” and “Without =”, in Figure 4. We also report the amount of performance gain explained by questions having an “=” vs. not in Appendix G.1. We find that the majority of the performance gain from CoT on MMLU and MMLU Pro comes from questions that have an “=” in the question or generated responses. Because “=” are usually found in math problems, we equate this to CoT primarily benefiting MMLU and MMLU Pro on the math-related questions with very little to no gain (depending on the model) for non-math questions.

G.1 PERFORMANCE IMPACTS OF “=” ON MMLU AND MMLU PRO

Tables 18 and 19 show the amount of total improvement from using CoT over direct prompting that can be explained by the presence of “=” on MMLU and MMLU Pro over multiple models.

H FULL RESULTS OF EVALUATIONS ON FORMAL REASONING DATASETS

As discussed in Section 5, we include detailed evaluation results of few-shot direct answer, few-shot CoT, direct answer solver, CoT solver, and tool-augmented prompting in Table 20. The unparseable rate stands for the rate of unparseable model responses that either fail to pass our answer extraction parser (for all methods except tool-augmented prompting) or fail to be executed by symbolic solvers. For FOLIO and ContextHub, we compute the accuracy by making a random guess for the unparseable responses; for GSM8K and GSM8K-Hard, we consider the unparseable responses as incorrect.

We note that all models have a low unparseable rate ($< 10\%$) for all methods except tool-augmented prompting. By manually inspecting the outputs, we observe that the high unparseable rate for some models with tool-augmented prompting is caused by these models generating Python programs or

⁶We explored implementing g with a logistic regression classifier with tf-idf features over the $(\mathbf{q}, \tilde{\mathbf{y}}_{cot})$ pairs, trained over a subset of the data from MMLU and MMLU Pro. This classifier actually allowed us to discover the “=” feature, but its accuracy did not exceed the accuracy of that single feature.

Table 18: Total CoT deltas on MMLU broken down by the total gain from questions and responses with an “=” vs. without an “=”.

Model	Total CoT Delta	CoT delta w/ =	CoT delta w/o =	Perf. Gain w/ =	Fraction of N w/ =
Llama 2 7b	6.0	0.6	5.4	9.8%	10.9%
Mistral 7b	4.1	1.2	2.9	28.6%	9.8%
Llama 3.1 8b	5.5	2.9	2.6	52.9%	9.6%
Llama 3.1 70b	1.9	1.8	0.1	94.0%	10.6%
Gemma 2 9b	2.6	2.0	0.6	78.5%	10.0%
Phi-3 Small 8k	3.1	1.5	1.7	47.4%	8.3%
Qwen 2 7b	2.5	3.0	-0.5	100.0%	9.8%
Qwen 2 72b	3.5	2.4	1.1	67.8%	9.6%
GPT-4o Mini	5.2	3.5	1.7	66.9%	10.5%
GPT-4o	4.2	2.4	1.8	57.6%	10.3%
Claude-3 Haiku	3.7	2.4	1.3	64.4%	9.3%
Claude-3.5 Sonnet	3.2	2.3	0.9	72.1%	10.7%
Gemini 1.5 Flash	3.0	1.7	1.2	59.0%	10.1%
Gemini 1.5 Pro	1.9	1.0	0.9	51.9%	9.6%

Table 19: Total CoT deltas on MMLU Pro broken down by the total gain from questions and responses with an “=” vs. without an “=”.

Model	Total CoT Delta	CoT delta w/ =	CoT delta w/o =	Perf. Gain w/ =	Fraction of N w/ =
Llama 2 7b	1.6	1.3	0.3	79.6%	43.6%
Mistral 7b	3.8	1.9	1.9	50.7%	41.8%
Llama 3.1 8b	12.4	10.0	2.4	80.8%	35.2%
Llama 3.1 70b	11.4	11.1	0.3	97.6%	39.6%
Gemma 2 9b	7.6	7.4	0.2	97.9%	40.2%
Phi-3 Small 8k	11.6	9.9	1.7	85.7%	42.7%
Qwen 2 7b	10.0	8.9	1.1	88.6%	41.6%
Qwen 2 72b	19.0	16.1	2.9	84.7%	41.4%
GPT-4o Mini	20.6	18.4	2.3	89.0%	44.0%
GPT-4o	17.7	17.1	0.6	96.7%	44.1%
Claude-3 Haiku	8.7	7.8	0.9	90.1%	42.0%
Claude-3.5 Sonnet	16.2	14.8	1.3	91.9%	43.4%
Gemini 1.5 Flash	12.9	11.8	1.1	91.3%	42.3%
Gemini 1.5 Pro	10.0	8.6	1.4	85.7%	41.8%